

Yuze Zhu

yuzezhu@berkeley.edu | [Website](#) | [GitHub](#) | [Google Scholar](#)

Education

University of California, Berkeley Berkeley, CA
Visiting Student in Computer Science, Fall 2026 *Aug. 2026 – Dec. 2026*

- Selected by UCAS through its Study Abroad Program (SAP); tuition funded by UCAS.

University of Chinese Academy of Sciences (UCAS) Beijing, China
B.Eng. in Artificial Intelligence *Sep. 2023 – Expected Jul. 2027*

- GPA: 3.87/4.00.
- Selected coursework: Computer Vision, Robotics, Pattern Recognition, Probability Theory, Natural Language Processing, and Data Structures.

Research Interests

My long-term goal is to build general-purpose robots that understand and interact with the physical world. I am particularly interested in **representation learning** and **spatial intelligence**, studying how robots can develop an intuitive understanding of 3D space to support robust generalization across tasks and environments.

Research Experience

University of California, Berkeley Berkeley, CA
Visiting Student | Advised by Dr. Roei Herzig (Darrell Group, BAIR) *Aug. 2026 – Present*

- Investigating learned representations in World Action Models (WAMs), with a focus on how task-relevant information emerges and supports generalization and adaptation in robotic manipulation.
- Exploring human-video in-context learning for robotic manipulation, studying how representations from demonstrations can be aligned with the robot’s current state and leveraged to guide future video and action generation.

FiveAges (Embodied AI Startup) & CASIA Beijing, China
Algorithm Intern | Advised by Prof. Yan Huang and Prof. Liang Wang *Mar. 2026 – Aug. 2026*

- Co-developed BridgeVLA++, a memory-augmented Vision-Language-Action framework for generalizable 3D manipulation, and led its simulation and real-robot evaluation; co-first-authored the manuscript submitted to IEEE TPAMI.

Institute of Automation, Chinese Academy of Sciences (CASIA) Beijing, China
Research Intern | Advised by Prof. Junyu Gao and Prof. Changsheng Xu *Jul. 2025 – Jan. 2026*

- Developed a closed-loop self-correction framework for continuous Vision-Language Navigation using world-model foresight and environmental feedback, and led the experimental implementation and evaluation on R2R-CE and RxR-CE; resulted in a second-author paper accepted at ICML 2026.

Selected Publications & Manuscripts

BridgeVLA++: A Data-Efficient, Generalizable, and Memory-Augmented Vision-Language-Action Framework for 3D Manipulation [Paper] [Project]

Peiyan Li*, Yuze Zhu*, Yixiang Chen, Qisen Ma, Yuan Xu, Jiabing Yang, He Guan, Yan Huang, Hongtao Wu, Xiao Ma, Tao Kong, Liang Wang, and Tieniu Tan

*Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. *Co-first authors.*

SC²-WM: A Self-Correcting World Model with Closed-Loop Feedback for Vision-and-Language Navigation in Continuous Environments [Paper]

Xuan Yao, Yuze Zhu, Junyu Gao, Zongmeng Wang, and Changsheng Xu

International Conference on Machine Learning (ICML), 2026.

M3D-VLA: A Memory-Augmented 3D Vision-Language-Action Model for History-Aware Manipulation

Yide Guan, Yuze Zhu, Peiyan Li, Jing Liu, Nianfeng Liu, and Yan Huang

Under review, 2026.

Service & Outreach

Chunfen Project Science Outreach Program

Core Volunteer

2023 – Present

- Delivered hands-on science lessons at 10+ primary schools in Beijing and contributed 60+ accessible science articles to the program’s online encyclopedia.

Skills

Robot Platforms: Franka Research 3, DOBOT CR5A

Programming: Python, PyTorch, C/C++

Languages: Chinese (Native); English (TOEFL 5.0/6.0, CEFR C1)